

Chapter:1 Binary Numbers; Boolean Algebra & logic gates

Number system

The numerical system in the system of naming or representing numbers.

BINARY NUMBER: (1,0) Base 2.

Decimal to binary ▶

given that $(15)_{10}$

$= (1111)_2$

Ans

$$\begin{array}{r} 2 \overline{) 15} \\ \underline{2 \times 7 = 14} \\ 1 \\ \underline{2 \times 3 = 6} \\ 1 \\ \underline{2 \times 1 = 2} \\ 1 \end{array}$$

$(41)_{10}$

$= (101001)_2$

$$\begin{array}{r} 2 \overline{) 41} \\ \underline{2 \times 20 = 40} \\ 1 \\ \underline{2 \times 10 = 20} \\ 0 \\ \underline{2 \times 5 = 10} \\ 0 \\ \underline{2 \times 2 = 4} \\ 1 \\ \underline{2 \times 1 = 2} \\ 0 \end{array}$$

Ex-1.3 $(0.6875)_{10}$ to binary

	int	+	frac
$0.6875 \times 2 =$	1		0.375
$0.375 \times 2 =$	0		+0.75
$0.75 \times 2 =$	1		+0.50
$0.50 \times 2 =$	1		+0.00

Co-eff

$$\begin{array}{c} 1 \\ 0 \\ 1 \\ 1 \end{array} \downarrow = (0.1011)_2$$

Ex 1.1

$(0.513)_{10}$ to Octal

$$0.513 \times 8 = 4.104$$

$$0.104 \times 8 = 0.832$$

$$0.832 \times 8 = 6.656$$

$$0.656 \times 8 = 5.248$$

$$0.248 \times 8 = 1.984$$

$$0.984 \times 8 = 7.872$$

... ..

$$(0.513)_{10} = (0.406517...)_{8}$$

OCTAL / HEXA

$$\left(\begin{array}{ccccccc} \underline{10} & \underline{110} & \underline{001} & \underline{101} & \underline{011} & \overset{(421)}{\underline{111}} & \underline{100} & \underline{000} & \underline{110} \\ 2 & 6 & 1 & 5 & 3 & 7 & 4 & 0 & 6 \end{array} \right)_2 = (26153.7406)_8$$

$$\left(\begin{array}{ccccccc} \underline{10} & \underline{1100} & \underline{0110} & \underline{1011} & \underline{1111} & \underline{0010} \\ 2 & C & 6 & B & F & 2 \end{array} \right)_2 = (2C6B.F2)_{16}$$

Boolean Algebra

x	y	x.y
0	0	0
0	1	0
1	0	0
1	1	1

AND

x	y	x+y
0	0	0
1	0	1
0	1	1
1	1	1

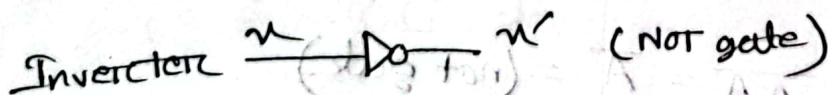
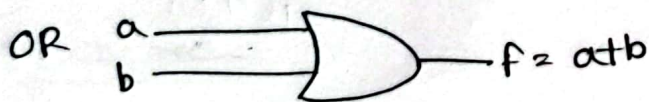
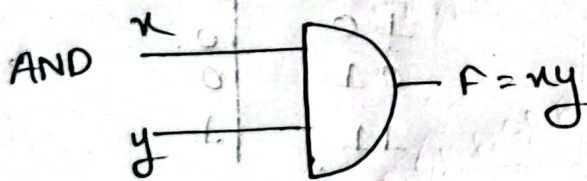
OR

x	x'
1	0
0	1

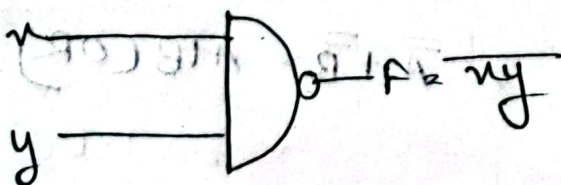
NOT

from this, $0+0=0$, $0+1=1$

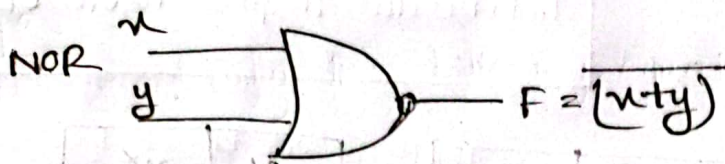
$1+1=1$; $1+0=0+1=1$



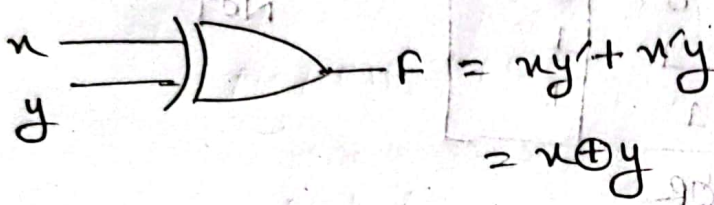
NAND



x	y	(xy)'
0	0	1
1	0	1
0	1	1
1	1	0



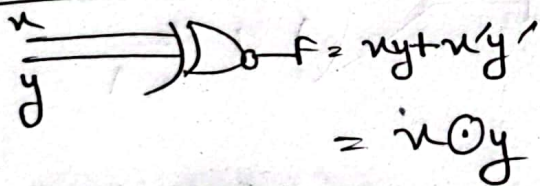
Exclusive-OR



x	y	$\overline{xy}$
0	0	1
0	1	0
1	0	0
1	1	0

x	y	$x \oplus y$
0	0	0
0	1	1
1	0	1
1	1	0

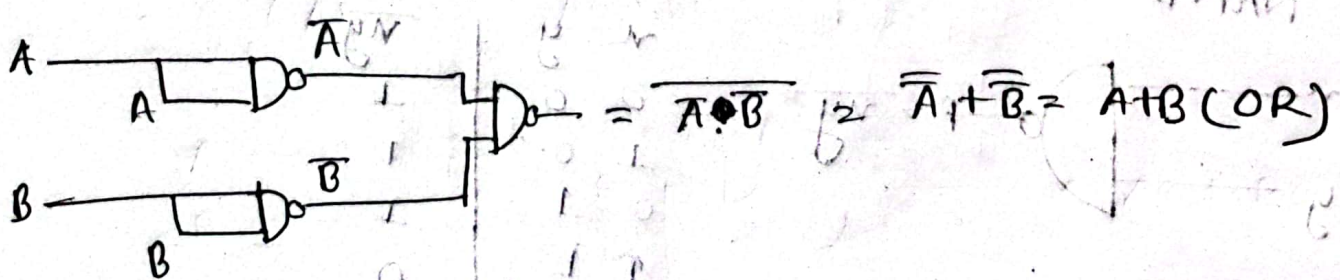
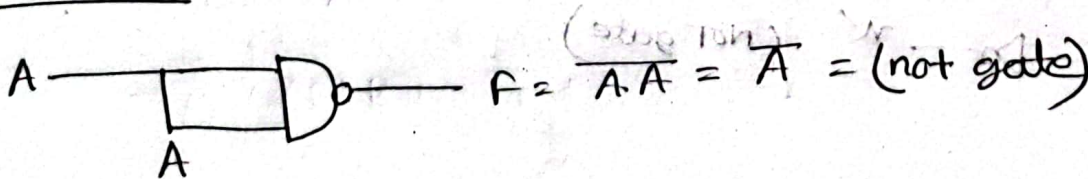
Ex-ANOR

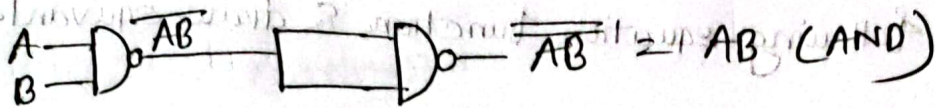


x	y	$\overline{x \oplus y}$
0	0	1
1	0	0
0	1	0
1	1	1

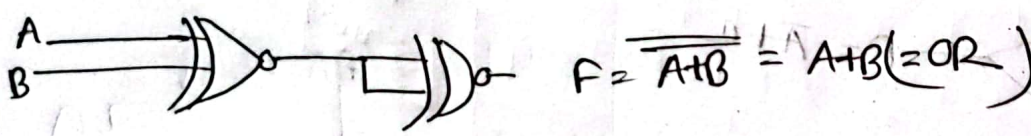
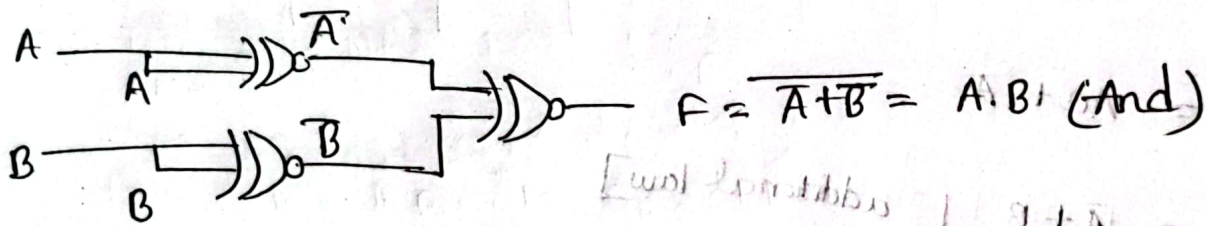
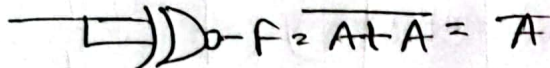
Design Basic gate by universal gate

NAND





NOR



Class-2

Date: 16/01/23

Distributive

$$A + BC = (A + B)(A + C)$$

$$A(B + C) = AB + AC$$

De Morgan's law

$$\overline{A+B} = \bar{A} \cdot \bar{B}$$

$$\overline{AB} = \bar{A} + \bar{B}$$

Absorption

$$A + AB = A$$

$$A(A + B) = A$$

Additional law

$$A + \bar{A} \cdot B = A + B$$

$$\bar{A} + A \cdot B = \bar{A} + B$$

[P-1]

Simplify the following equation function & draw equivalent

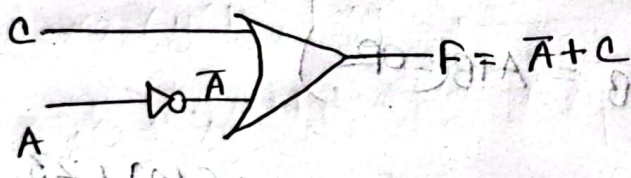
ckt:-

$$F = ABC + AB\bar{C} + \bar{A}$$

$$= AC(B + \bar{B}) + \bar{A}$$

$$= AC + \bar{A}$$

$$= \bar{A} + B \quad [\text{additional law}]$$



[P-2]

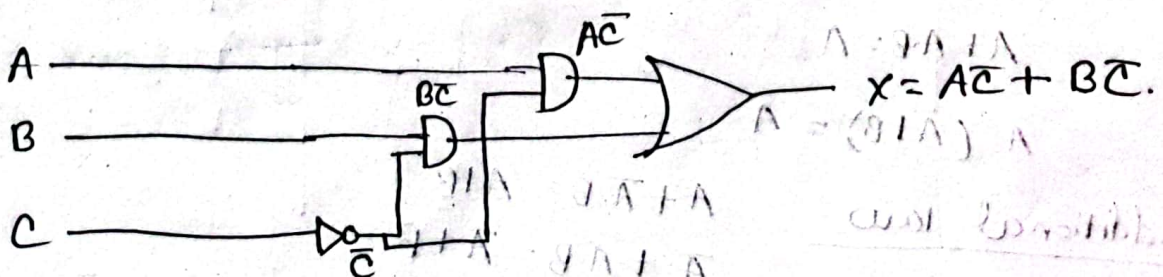
$$\bar{A}B\bar{C} + A\bar{B}\bar{C} + A\bar{B}C$$

$$= \bar{B}\bar{C}(A + \bar{A}) + A\bar{B}C$$

$$= \bar{B}\bar{C} + A\bar{B}C$$

$$= \bar{C}(A + \bar{B})$$

$$= \bar{C}(A + \bar{B}) = A\bar{C} + \bar{B}\bar{C}$$



P-3 $(A+B)(A+B+D)\bar{D}$

$$= (\bar{A}\bar{D} + B\bar{D})(\bar{A}\bar{D} + B\bar{D} + D\bar{D})$$

$$= (\bar{A}\bar{D} + B\bar{D})(\bar{A}\bar{D} + B\bar{D})$$

$$= (\bar{A}+B)(\bar{A}+B)\bar{D}$$

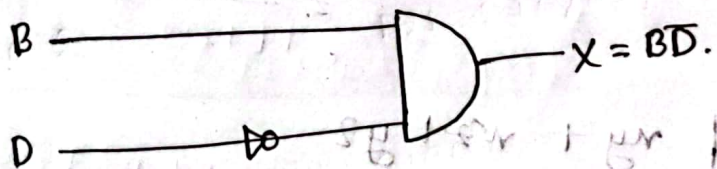
$$= (\bar{A}\bar{A} + \bar{A}B + A\bar{B} + BB)\bar{D}$$

$$= (\bar{A}B + B(A+\bar{A}))\bar{D}$$

$$= (\bar{A}B + AB)\bar{D}$$

$$= B(A+\bar{A})\bar{D}$$

$$= B\bar{D}$$



P-4 $x + x'y$

$$= (x + x')(x + y)$$

$$= 1(x + y)$$

$$= x + y$$

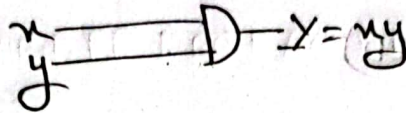


P-5 $x(x+y)$

$$= xx + xy$$

$$= 0 + xy$$

$$= xy$$

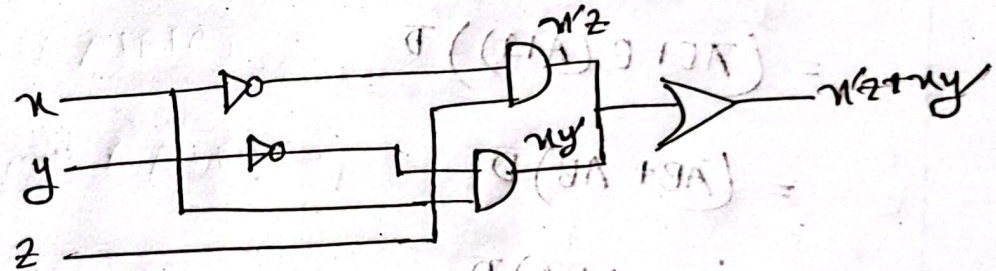


P-6

$$x'y'z + x'yz + xy'$$

$$= x'z(y'+y) + xy'$$

$$= x'z + xy'$$



P-7 $xy + x'z + yz$

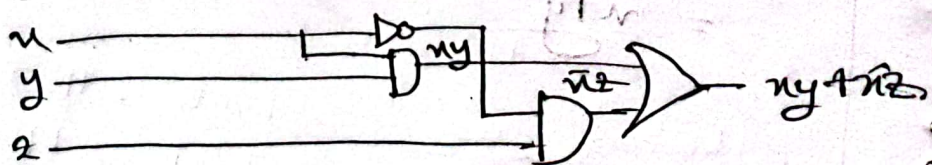
$$= xy + x'z + yz(x+x')$$

$$= xy + x'z + xy'z + x'yz$$

$$= xy(1+z) + x'z(1+y)$$

$$= xy + yz + x'z + x'y$$

~~xy + yz~~



$\Sigma(1)$ $\Pi(0)$
Minterm / Maxterm

	A	B	C	Minterm	Maxterm	
0 →	0	0	0	$\bar{A}\bar{B}\bar{C}$	$A+B+C$	m_0
1 →	0	0	1	$\bar{A}\bar{B}C$	$A+B+\bar{C}$	m_1
2 →	0	1	0	$\bar{A}B\bar{C}$	$A+\bar{B}+C$	m_2
3 →	0	1	1	$\bar{A}BC$	$A+\bar{B}+\bar{C}$	m_3
4 →	1	0	0	$A\bar{B}\bar{C}$	$\bar{A}+B+C$	m_4
5 →	1	0	1	$A\bar{B}C$	$\bar{A}+B+\bar{C}$	m_5
6 →	1	1	0	$AB\bar{C}$	$\bar{A}+\bar{B}+C$	m_6
7 →	1	1	1	ABC	$\bar{A}+\bar{B}+\bar{C}$	m_7

$$F(A,B,C) = \Sigma(0,1,2,3,4,5,6,7)$$

$$= (\text{Sum of products})$$

$$= \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}B\bar{C} + \bar{A}BC + A\bar{B}\bar{C} + A\bar{B}C + AB\bar{C} + ABC$$

$$F(A,B,C) = \Pi(2,4,6)$$

$$= (\text{Product of sums})$$

$$= (A+B+C) \cdot (\bar{A}+B+C) \cdot (\bar{A}+\bar{B}+C)$$

[P-8] $F(A, B, C) = \pi(1, 3, 5)$

$$= (A+B+C) \cdot (A+B+C) \cdot (A+B+C)$$

$$= (AA + \bar{A}\bar{B} + A\bar{C} + \bar{A}B + \bar{B}\bar{B} + B\bar{C} + A\bar{C} + \bar{B}\bar{C} + \bar{C}\bar{C}) \cdot (A+B+C)$$

$$= (A + A(B+B) + A\bar{C} + \bar{C}(B+B)) \cdot (A+B+C)$$

$$= (A + A + A\bar{C} + \bar{C}) \cdot (A+B+C)$$

$$= (A + A\bar{C} + \bar{C}) \cdot (A+B+C)$$

$$= (A(1+\bar{C}) + \bar{C}) \cdot (A+B+C)$$

$$= (A + \bar{C}) \cdot (A+B+C)$$

$$= \underline{A\bar{A}} + AB + A\bar{C} + \bar{A}\bar{C} + B\bar{C} + \bar{C}\bar{C}$$

$$= AB + A\bar{C} + \bar{A}\bar{C} + B\bar{C}$$

$$= AB + \bar{C}(A + \bar{A}) + B\bar{C}$$

$$= AB + \bar{C} + B\bar{C}$$

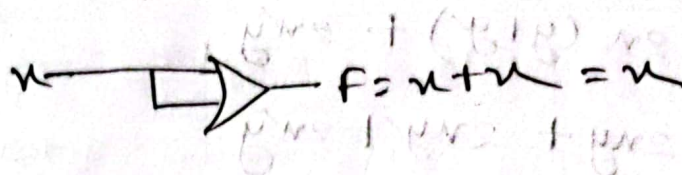
$$= AB + \bar{C}(1+B)$$

$$= AB + \bar{C}$$

P-9 $xy + xy'$

$$= x(y + y')$$

$$= x \cdot 1$$



P-10 $(x + y)(x + y')$

$$= x + xy + xy' + yy'$$

$$= x + x(y + y') + 0$$

$$= x + x \cdot 1$$

$$= x + x$$

$$= x$$



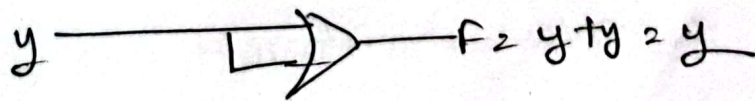
P-11 $xy + x'y + xy'z$

$$= xy(z + z') + x'y$$

$$= xy + x'y$$

$$= y(x + x')$$

$$= y \cdot 1 = y$$



P-12

$$\textcircled{1} \quad 2x + 2xy$$

$$= 2x(y + y') + 2xy$$

$$= 2xy + 2xy' + 2xy$$

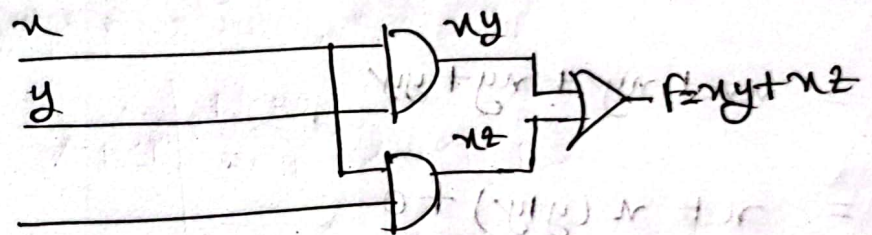
$$= 2y(x + x') + 2xy'$$

$$= 2y + 2xy'$$

$$= 2(y + xy')$$

$$= 2(x + y)$$

$$= (xy + xz)$$



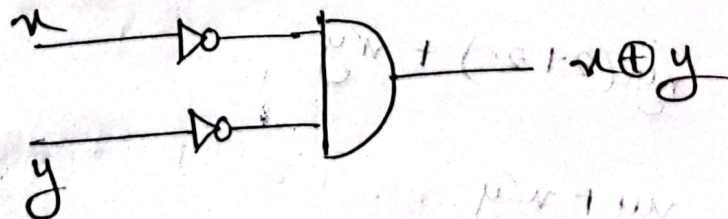
P-13 $(A+B)(A'+B')$

$$= (\underline{A}A' + AB' + A'B + \underline{B}B')$$

$$= (AB' + A'B)$$

$$= \overline{(AB)(A'B)}$$

$$= A \oplus B$$



P-14

$$F = ABC + A'BC + ABC + A'B'C'$$

$$= AC(B' + B) + \cancel{A(B'B)} A'BC + A'B'C'$$

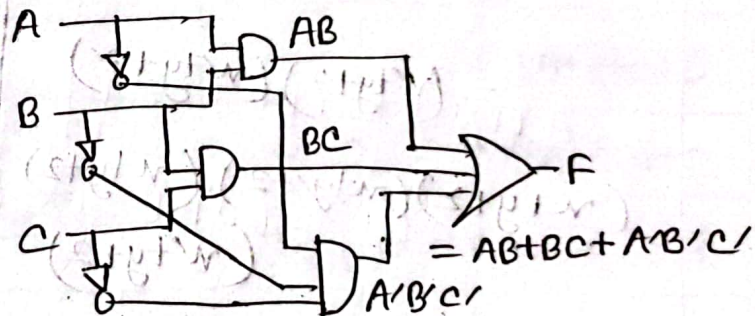
$$= AC + A'BC + A'B'C'$$

$$= C(A + A'B) + A'B'C'$$

$$= C(A + A')(A + B) + A'B'C'$$

$$= C(A + B) + A'B'C'$$

$$= AC + BC + A'B'C'$$



P-15 $F = A + B'C$

$$= AC(B + B') + B'C(A + A')$$

$$= AB + AB' + AB'C + A'B'C'$$

$$= (C + C')AB + AB'(C + C') + AB'C + A'B'C'$$

$$= ABC + ABC' + AB'C + AB'C' + AB'C + A'B'C'$$

$$= m_7 + m_6 + m_5 + m_4 + m_2$$

$$= m_1 + m_4 + m_5 + m_6 + m_7$$

$$= \sum (1, 2, 5, 6, 7) \quad (\text{Sum of minterms})$$

P-16 $F = xy + xz$

$$= (xy + xz) (xy + xz)$$

$$= (x + x) (x + y) (x + z) (y + z)$$

$$= (x + y) (x + z) (y + z)$$

$$= (x + y + z) (x + y + z)$$

$$(x + y + z) (x + y + z) (x + y + z)$$

$$= m_4 \cdot m_5 \cdot m_0 \cdot m_2$$

$$= m_0 m_2 m_4 m_5$$

$$= \prod (0, 2, 4, 5) \text{ (product of Sum)}$$

Now,

$$x + y = x + y + z + z$$

$$= (x + y + z) (x + y + z)$$

$$x + z = x + z + y + y$$

$$= (x + z + y) (x + z + y)$$

$$y + z = y + z + x + x$$

$$= (y + z + x) (y + z + x)$$

P-17 Max & min terms for 4 inputs

A	B	C	D	min	max
0	0	0	0	$\bar{A}\bar{B}\bar{C}\bar{D}$	$A+B+C+D$
0	0	0	1	$\bar{A}\bar{B}\bar{C}D$	$A+B+C+\bar{D}$
0	0	1	0	$\bar{A}\bar{B}C\bar{D}$	$A+B\bar{C}+D$
0	0	1	1	$\bar{A}\bar{B}CD$	$A+B\bar{C}+\bar{D}$
0	1	0	0	$\bar{A}B\bar{C}\bar{D}$	$A+\bar{B}+C+D$
0	1	0	1	$\bar{A}B\bar{C}D$	$A+\bar{B}+C+\bar{D}$
0	1	1	0	$\bar{A}BC\bar{D}$	$A+\bar{B}+\bar{C}+D$
0	1	1	1	$\bar{A}BCD$	$A+\bar{B}+\bar{C}+\bar{D}$
1	0	0	0	$A\bar{B}\bar{C}\bar{D}$	$\bar{A}+B+C+D$
1	0	0	1	$A\bar{B}\bar{C}D$	$\bar{A}+B+C+\bar{D}$
1	0	1	0	$A\bar{B}C\bar{D}$	$\bar{A}+B\bar{C}+D$
1	0	1	1	$A\bar{B}CD$	$\bar{A}+B\bar{C}+\bar{D}$
1	1	0	0	$AB\bar{C}\bar{D}$	$\bar{A}+\bar{B}+C+D$
1	1	0	1	$AB\bar{C}D$	$\bar{A}+\bar{B}+C+\bar{D}$
1	1	1	0	$ABC\bar{D}$	$\bar{A}+\bar{B}+\bar{C}+D$
1	1	1	1	$ABCD$	$\bar{A}+\bar{B}+\bar{C}+\bar{D}$

P-18 $F(A,B,C,D) = D(A+B) + B'D$. Express in sum of minterms & maxterms.

$$= AD + BD + B'D$$

$$= AD(B+B') + BD(A+A') + B'D(A+A')$$

$$= A'BD + A'B'D + ABD + A'BD + AB'D + A'B'D$$

$$= ABD(C+C') + (A'B'D)(C+C') + ABD(A+C) + A'BD(C+C')$$

$$AB'D(C+C') + (A'B'D)(C+C')$$

$$= A'BDC + A'BDC' + A'B'DC + A'B'DC' + ABDC + AB'C'D + A'BCD$$

$$+ A'BC'D + AB'C'D + A'B'C'D + A'B'C'D$$

$$= \cancel{m_0} + m_7 + m_5 + m_3 + m_1 + m_{15} + m_{13} + m_{15}$$

$$= \sum (1, 3, 5, 7, 9, 11, 13, 15)$$

$$= \prod (0, 2, 4, 6, 8, 10, 12, 14)$$

Ans

P-19 $F = [(AB)A] [(AB)B]$ QAM-X

Complement of F and simplify F' .

$$F' = ([(AB)A] [(AB)B])'$$

$$= [(AB)A]' + [(AB)B]'$$

$$= (AB + A') (AB + B')$$

$$= AB + A'B + AB + AB'$$

$$= AB + A'B + AB'$$

$$= A'B + AB'$$

$$= A' + 1$$

$$= 1$$

$\therefore F' = 1$

P-20

$$(A+B+C)'$$

$$= (A+W)'$$

Let, $W = B+C$

$$= A'W'$$

$$= A' (B+C)'$$

$$= A' (B' \cdot C')$$

$$= A'B'C' \quad \underline{\text{Ans}}$$

K-MAP

Kmap is used to simplify the boolean expression in pictorial format without using boolean laws and theorems.

Developed by karnaugh (1953).

Formula: 2^n where n is the no. of variable.

Ex: $n=2$

A \ B	0	1
0	0	1
1	3	2

$2^2 = 4 \text{ cell}$

$n=3$

A \ BC	00	01	11	10
0	0	1	3	2
1	4	5	7	6

$2^3 = 8 \text{ cell}$

$n=4$

AB \ CD	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

$2^4 = 16 \text{ cell}$

Rule:-

- (i) No zero allowed
- (ii) Group can be vertical or horizontal, cannot diagonal
- (iii) Overlapping allowed
- (iv) Group must be large as possible
- (v) Group must contain 2^n cells

KARNAUGH MAP (K-MAP)

① Pair group: 2 cells

1		1	
1		1	

② Quad group: 4 cells

	1	1	
1	1	1	1
1	1	1	1
	1	1	

③ Octal group: 8 cells

Ex: 1 $F(A,B) = \sum (0, 2, 3)$

	B	0	1
A	0	1	
	1	1	

X	0	0	X
0	X		
		0	0
0		0	0

Don't Care:

P-1

$$F(A, B, C, D) = \sum (0, 1, 2, 3, 5, 7, 9, 11, 13) + d(4, 6, 8, 10, 12)$$

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	1	1	1	1
$\bar{A}B$	x	1	1	x
AB	x	1		
$A\bar{B}$	x	1	1	x

$$= \bar{A} + \bar{B} + \bar{C}$$

P-2 $F(A, B, C, D) = \prod (1, 3, 5, 6, 8, 9, 10, 12, 13) + d(0, 2, 7)$

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	x	0	0	x
$\bar{A}B$			x	0
AB	0	0		
$A\bar{B}$	0	0		0

⊛ Incomplete

NAND & NOR IMPLEMENTATION:

$$F(A, B, C) = \sum (0, 3, 7)$$

$$= \bar{A}\bar{B}\bar{C} + \bar{A}BC + ABC$$

(i) Sum of product [SOP] implement by NAND gate.

(ii) if we want to solve it by NOR then we have to

follow the process,

SOP $\rightarrow$ POS $\rightarrow$ NOR

$$F(A, B, C) = \pi(0, 3, 7)$$

(i) Product of sum [POS] implement by NOR gate.

(ii) if we want to solve it by NAND then we have to

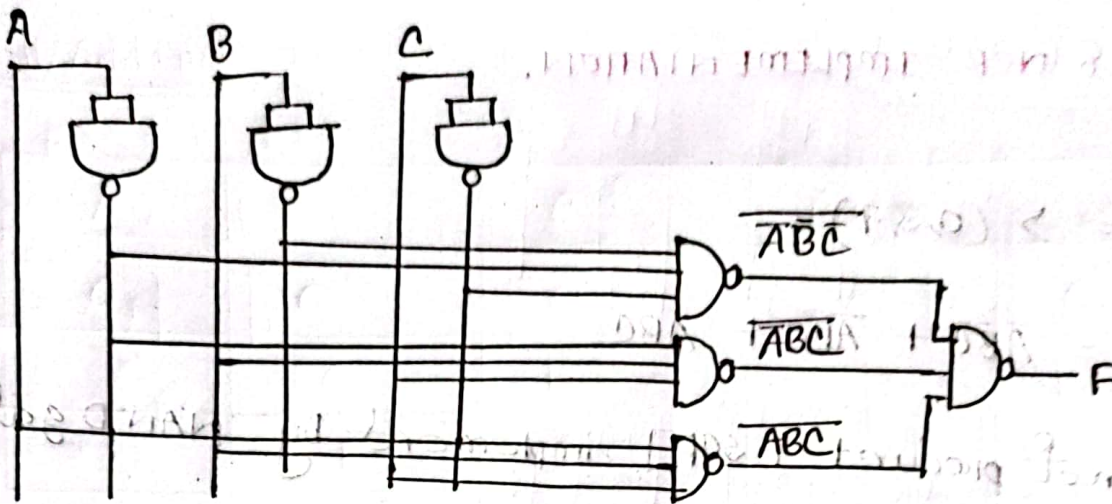
follow the process,

POS $\rightarrow$ SOP $\rightarrow$ NAND.

problem-1

$$F(A, B, C) = \pi(0, 3, 7)$$

$$= \bar{A}\bar{B}\bar{C} + \bar{A}BC + ABC$$



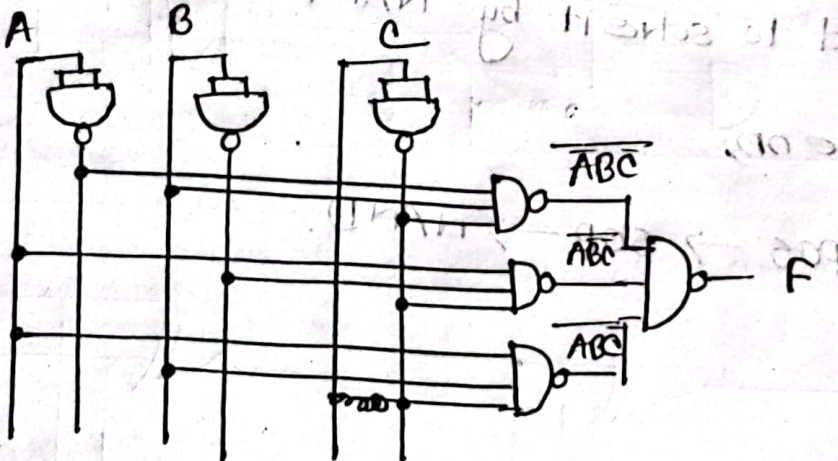
$$F = \overline{A} \cdot \overline{B} \cdot \overline{C} + \overline{A} \cdot \overline{B} \cdot C + \overline{A} \cdot B \cdot C$$

$$= \overline{A \cdot B \cdot C} + \overline{A \cdot B \cdot C} + \overline{A \cdot B \cdot C}$$

$$= \overline{A \cdot B \cdot C} + \overline{A \cdot B \cdot C} + \overline{A \cdot B \cdot C}$$

Q. $F(A, B, C) = \sum(2, 4, 6)$

$$F = \overline{A}BC + A\overline{B}C + ABC$$



$$F = \overline{A}BC + A\overline{B}C + ABC$$

$$= \overline{A}BC + A\overline{B}C + ABC$$

$$= \overline{A}BC + A\overline{B}C + ABC$$

Half Adder

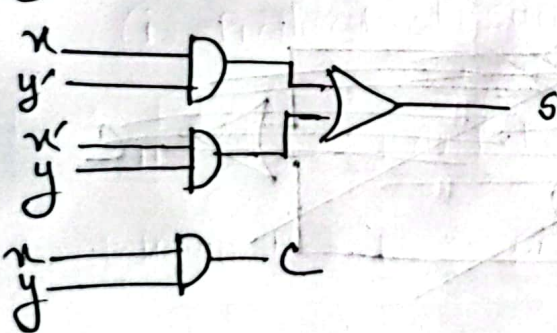
x	y	c	s
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

$$\therefore s = xy' + xy$$

$$= x \oplus y \quad [\text{X-OR gate}]$$

$$c = xy \quad (\text{AND gate output})$$

(a)



(b)

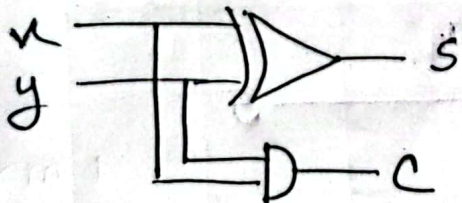
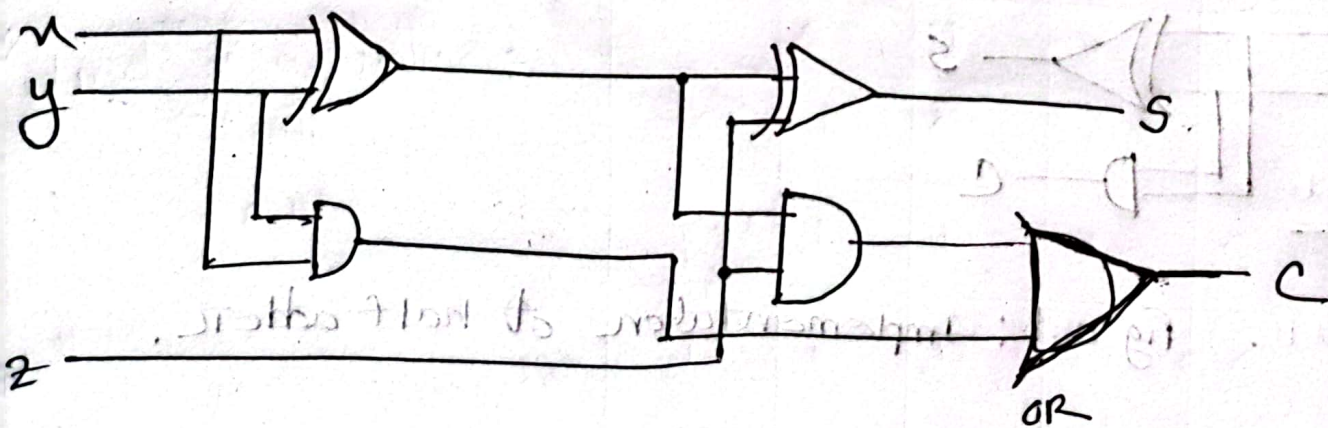
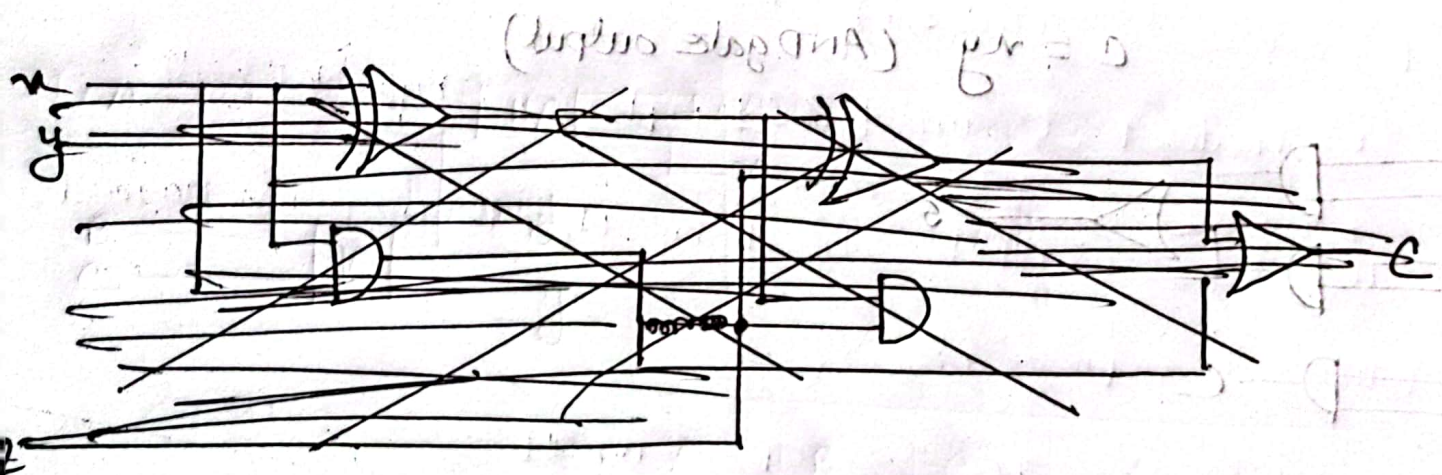


fig a.b: Implementation of half adder.

Full Adder

A	B	C	Carry	Sum
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1



$$S = z \oplus (x \oplus y)$$

$$= z'(xy' + x'y) + z(xy' + x'y)$$

$$= xy'z' + x'yz' + xy'z + x'yz$$

$$C = z(xy' + x'y) + xy = xy'z + x'yz + xy$$

Eqn for carry:

$$\text{Carry}(A, B, C) = \sum (3, 5, 6, 7)$$

$$= \bar{A}BC + A\bar{B}C + AB\bar{C} + ABC$$

	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
$\bar{A}$			1	
A		1	1	1

$$= BC + AC + AB$$

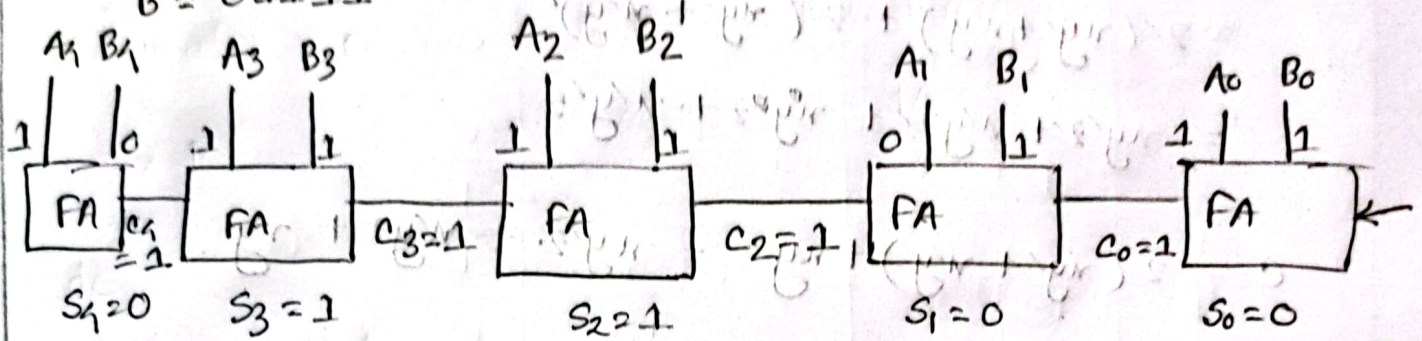
$$\text{Sum}(A, B, C) = \sum (1, 2, 4, 7)$$

$$= \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC$$

	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
$\bar{A}$		1		1
A	1		1	

Q. $A = 11101$

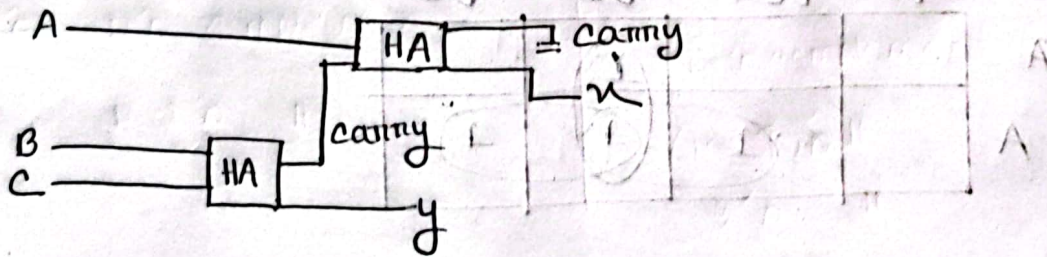
$B = 01111$



$$= C_4 S_4 S_3 S_2 S_1 S_0 = 101100$$

Ans

Ques:-



find n, y value:-

A	B	C	n	y
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	1	1	0	0

(A-B)

A	B	B (Borrow)	Sum
0	0	0	0
0	1	1	0
1	0	0	1
1	1	0	0

Full Subtractor:

$$A - B - C$$
$$= A \oplus (B \oplus C)$$

A	B	C	B	Sum
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	0
1	0	0	0	1
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

Number System

Decimal Binary
Base $\rightarrow$ 10 2

Octal	Hexa-decimal
8	16

Code

(i) Excess 3-code

(ii) 8421 code

(iii) 2421 code

(iv) Gray code

(*) 1's complement

$n-1$'s

$n=8$ (Octal)

8's complement

$8-1$'s

7's

decimal

10's complement

$(10-1)$'s

9's

$n=2$ base / index

Ex

-17 show in binary using complement